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# Fast Fourier Transform-Based Distribution of Relaxation Times Analysis for Efficient and Flexible Time-Domain Electrochemical Impedance Characterization

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Electrochemical impedance spectroscopy (EIS) is a powerful analytical technique offering broad frequency range and straightforward implementation. However, low-frequency measurements are constrained by lengthy acquisition times and potential disruption of steady-state conditions. While the distribution of relaxation times (DRT) can accelerate impedance acquisition and interpret EIS data, current approaches are limited to pulse signals and their linear combinations. Herein, we present a novel, fast Fourier transform-based DRT formulation for rapid EIS acquisition with arbitrary signals and DRT deconvolution. Our approach demonstrates computational efficiency and improved DRT recovery, advancing opportunities for fast, efficient EIS characterization.

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Electrochemical impedance spectroscopy (EIS) has emerged as a powerful characterization technique with applications spanning energy[1,2] and medicine.[3] The widespread adoption of EIS can be attributed to its non-invasive nature, operational simplicity, and remarkable ability to probe electrochemical processes across a broad frequency range (from mHz to MHz).[4] However, measurements at low frequencies (<1 Hz) present significant practical challenges due to their inherently long duration, as the experimental time scales inversely with the sampled frequencies. This extended measurement period poses particular challenges when characterizing closed systems. A notable example is battery characterization during charging and discharging cycles, where applying sinusoidal current inputs at low frequencies inevitably shifts the state of charge during the measurement, thereby violating the steady-state conditions needed for EIS acquisition.[5]

To overcome the limitations inherent in conventional sinusoidal EIS measurements, researchers have employed the distribution of relaxation times (DRT) method, utilizing current pulses[6,7] and their linear combinations[8] as input signals. This pulse-based DRT approach offers two key advantages: 1) accelerated data acquisition compared to traditional sinusoidal excitation methods,[9] and 2) simplified instrumentation requirements.[10] The use of pulse signals for time-domain EIS acquisition[7,8,11] stems from the fact that, under the relaxation assumption, the corresponding voltage response has a closed-form expression.[6] In contrast to conventional methods that rely on sequential sinusoidal signal generation and acquisition, or utilize the fast Fourier transform (FFT) to directly calculate impedance from the voltage-to-current ratio,[12] the DRT-based method initially derives the DRT from the measured voltage response originating from the current pulse. It then leverages this derived DRT distribution as a constraint to estimate the impedance.[7]

To overcome these limitations, we introduce a novel FFT-based, DRT-constrained acquisition method capable of handling arbitrary input currents. This approach not only enhances DRT recovery and impedance reconstruction compared to existing pulse-based techniques,[6] but also significantly reduces computational time. Specifically, when applied to pseudorandom binary sequence (PRBS) signals, our method achieved a 27-fold reduction in processing time compared to traditional time-domain pulse DRT analysis.[6] Furthermore, the versatility of our method extends beyond pulse-derived waveforms, as demonstrated through validation with continuous and discontinuous DRTs, and on synthetic battery data.

## Methods

***Theory.***—We consider an input current, $i(t)$, with its Fourier transform defined as[13]

$$I(\omega) = \mathscr{F}[i(t)](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} i(t) e^{-j\omega t}\, \mathrm{d}t \qquad [1]$$

and inverse transform

$$i(t) = \mathscr{F}^{-1}[I(\omega)](t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} I(\omega) e^{j\omega t}\, \mathrm{d}\omega \qquad [2]$$

The voltage response $v(t)$ to the input current $i(t)$ is given by

$$v(t) = \mathscr{F}^{-1}[Z(\omega) I(\omega)](t) \qquad [3]$$

where $Z(\omega)$ is the impedance of the electrochemical system under study. The DRT model represents the impedance as[2,14]

$$Z_{\mathrm{DRT}}(\omega) = R_\infty + \int_{-\infty}^{+\infty} \frac{\gamma(\log\tau)}{1 + j\omega\tau}\, d\log\tau \qquad [4]$$

where $Z_{\mathrm{DRT}}(\omega)$ is the DRT-modeled impedance at the angular frequency $\omega$, $R_\infty$ is the ohmic resistance, $\tau$ is a timescale, and $\gamma(\log\tau)$ is the DRT. When substituting (4) into (3), we obtain

$$v(t) = R_\infty i(t) + \mathscr{F}^{-1}\left[\int_{-\infty}^{+\infty} \frac{1}{1 + j\omega\tau} I(\omega)\, \gamma(\log\tau)\, \mathrm{d}\log\tau\right](t) \qquad [5]$$

By applying the linearity of the Fourier transform, this expression can be recast as

$$v(t) = R_\infty i(t) + \int_{-\infty}^{+\infty} \mathscr{F}^{-1}\left[\frac{1}{1 + j\omega\tau} I(\omega)\right](t)\, \gamma(\log\tau)\, \mathrm{d}\log\tau \qquad [6]$$

To numerically approximate (6), we can sample $\gamma(\log\tau)$ at $N$ logarithmically equispaced timescales such that $\log\tau_1 < \log\tau_2 < \ldots < \log\tau_N$

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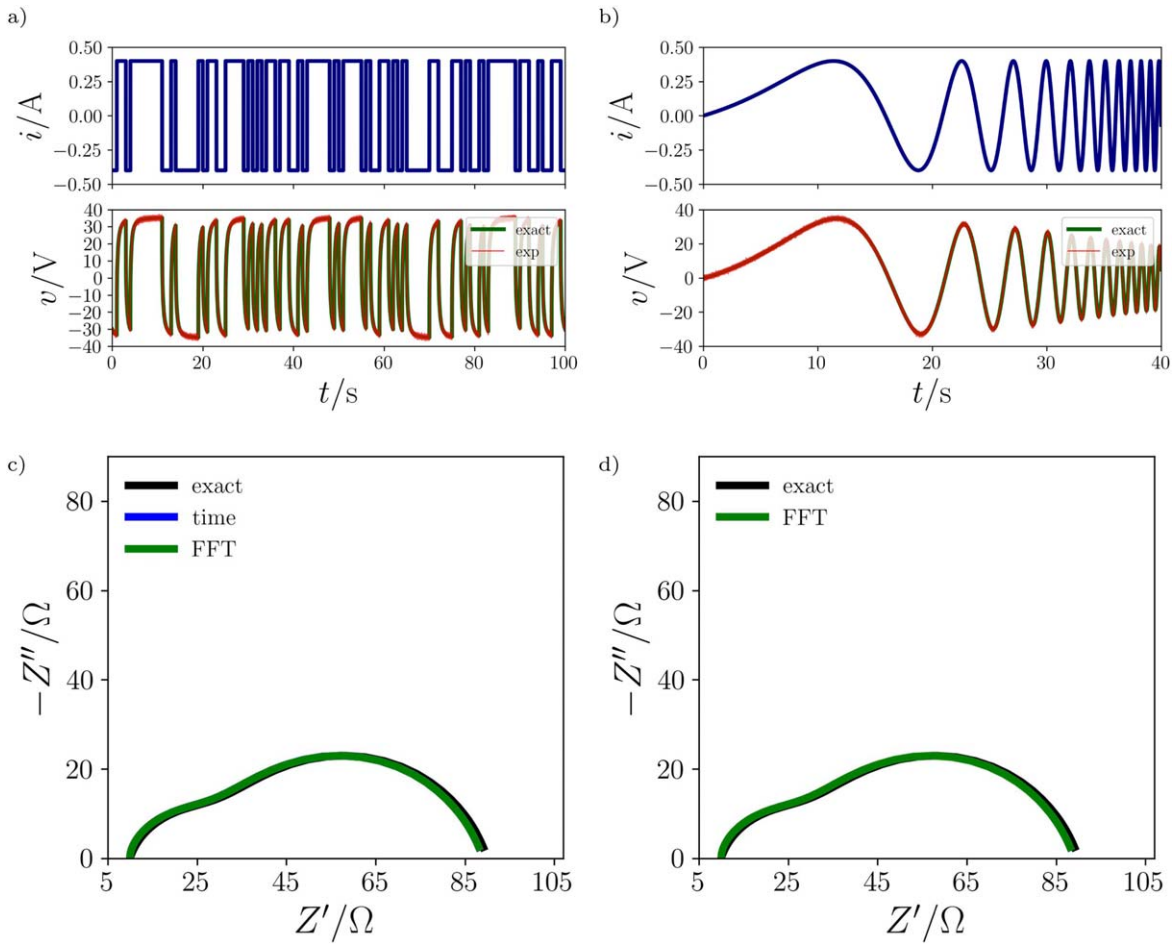


**Figure 1.** Method validation using synthetic experiments with the 2 × ZARC model. Input signals shown: (a) PRBS and (b) logarithmic chirp current-voltage pairs. Panels (c) and (d) present the corresponding Nyquist plots comparing recovered impedances against exact solutions.

and $\Delta \log \tau = \log \tau_{n+1} - \log \tau_n$ for $n = 1, 2, \ldots, N-1$. Applying the midpoint rule yields

$$v(t) = R_\infty i(t) + \sum_{n=1}^{N} \mathscr{F}^{-1}\left[\frac{1}{1 + j\omega\tau_n}\mathscr{F}[i(t)](\omega)\right] (t)\, \gamma(\log \tau_n)\, \Delta \log \tau \quad [7]$$

For $M$ times sampled, $t_1 < t_2 < \ldots < t_M$ with $t_{m+1} - t_m = \Delta t$ for $m = 1, 2, \ldots M-1$, (7) can be expressed in matrix form as

$$\mathbf{v} = \mathbf{B}_{\mathrm{FFT}}\mathbf{x} \quad [8]$$

where $\mathbf{v} = (v(t_1), v(t_2), \ldots, v(t_M))^\top$, $\mathbf{x} = (R_\infty, \gamma(\log \tau_1), \gamma(\log \tau_2), \ldots, \gamma(\log \tau_N))^\top$, and the entries of the matrix $(\mathbf{B}_{\mathrm{FFT}})_{m,n}$ for $m = 1, 2, \ldots, M$ and $n = 1, 2, \ldots, N+1$ are

$$(\mathbf{B}_{\mathrm{FFT}})_{m,n} = \begin{cases} i(t_m) & \text{if } n = 1 \\ \mathrm{FFT}^{-1}\left[\dfrac{1}{1 + j\boldsymbol{\omega}\tau_{n-1}}\mathrm{FFT}[i(\mathbf{t})](\boldsymbol{\omega})\right](t_m)\, \Delta \log \tau & \text{if } n > 1 \end{cases} \quad [9]$$

where FFT and $\mathrm{FFT}^{-1}$ denote the fast Fourier Transform and its inverse, respectively, $\mathbf{t} = (t_1, t_2, \ldots, t_M)^\top$ is the vector of time data, and the angular frequency vector is obtained by FFTing the time data, *i.e.*, $\boldsymbol{\omega} = 2\pi/(M\Delta t)(0, 1, \ldots, M/2 - 1, -M/2, \ldots, -1)^\top$ if $M$ is even, and $\boldsymbol{\omega} = 2\pi/(M\Delta t)(0, 1, \ldots, (M-1)/2, -(M-1)/2, \ldots, -1)^\top$ if $M$ is odd.

In the step method,[6] a rectangular pulse input of amplitude $\Delta i$ and duration $T_{\mathrm{pulse}}$ is considered with

$$i(t) = \Delta i(H(t) - H(t - T_{\mathrm{pulse}})) \quad [10]$$

where $H(\cdot)$ is the Heaviside function. The corresponding voltage response, $v_{\mathrm{pulse}}(t)$, is a special case of (6), and is given by

$$v_{\mathrm{pulse}}(t) = R_\infty\, i(t) + \Delta i \int_{-\infty}^{+\infty} [H(t)(1 - e^{-\frac{t}{\tau}}) - H(t - T_{\mathrm{pulse}}) \times (1 - e^{-\frac{t - T_{\mathrm{pulse}}}{\tau}})]\, \gamma(\log \tau)\, \mathrm{d} \log \tau \quad [11]$$

As above, the midpoint rule yields a discretization matrix $\mathbf{B}_{\mathrm{time}}$ given by

$$(\mathbf{B}_{\mathrm{time}})_{m,n} = \begin{cases} i(t_m) & \text{if } n = 1 \\ \Delta i\left[H(t_m)\left(1 - e^{-\frac{t_m}{\tau_{n-1}}}\right) - H(t_m - T_{\mathrm{pulse}})\left(1 - e^{-\frac{t_m - T_{\mathrm{pulse}}}{\tau_{n-1}}}\right)\right]\Delta \log \tau & \text{if } n > 1 \end{cases} \quad [12]$$

**Table I. Performance comparison of time-domain and FFT methods across 100 synthetic experiments using the 1 × ZARC, 2 × ZARC, and PWC models. Metrics include normalized residuals for DRT and impedance calculations, and computational time for each input current, which was normalized with respect to the lowest value obtained with the FFT method for each impedance model. The lowest value in each row is highlighted in bold.**

1 × ZARC

| Current shape | $\varepsilon_\gamma \times 10^{-2}$ | | $\varepsilon_Z \times 10^{-3}$ | | Normalized clock time | |
|---|---|---|---|---|---|---|
| | Time | FFT | Time | FFT | Time | FFT |
| Pulse | 0.135 ± 0.079 | 0.126 ± 0.078 | 0.055 ± 0.010 | 0.058 ± 0.011 | **0.82 ± 0.0019** | 1.09 ± 0.0017 |
| PRBS | **0.060 ± 0.025** | **0.048 ± 0.025** | **0.051 ± 0.003** | 0.053 ± 0.003 | 32.2 ± 0.0015 | 1.14 ± 0.0017 |
| Sawtooth | — | 0.159 ± 0.097 | — | 0.072 ± 0.022 | — | **1.00 ± 0.0017** |
| Linear chirp | — | 0.533 ± 0.247 | — | 0.054 ± 0.009 | — | 1.15 ± 0.0008 |
| Quadratic chirp | — | 0.167 ± 0.076 | — | 0.051 ± 0.004 | — | 1.12 ± 0.0010 |
| Logarithmic chirp | — | 0.0657 ± 0.035 | — | **0.051 ± 0.003** | — | 1.08 ± 0.0008 |
| Hyperbolic chirp | — | 0.111 ± 0.068 | — | 0.055 ± 0.008 | — | 1.03 ± 0.0015 |

2 × ZARC

| Current shape | $\varepsilon_\gamma \times 10^{-2}$ | | $\varepsilon_Z \times 10^{-3}$ | | Normalized clock time | |
|---|---|---|---|---|---|---|
| | Time | FFT | Time | FFT | Time | FFT |
| Pulse | 0.132 ± 0.065 | 0.123 ± 0.058 | 0.30 ± 0.03 | 0.31 ± 0.03 | **0.725 ± 0.0019** | 1.01 ± 0.0018 |
| PRBS | **0.070 ± 0.018** | **0.052 ± 0.018** | **0.29 ± 0.011** | 0.29 ± 0.013 | 7.7 ± 0.0015 | 1.01 ± 0.0012 |
| Sawtooth | — | 0.353 ± 0.197 | — | 0.32 ± 0.06 | — | 1.30 ± 0.0018 |
| Linear chirp | — | 0.543 ± 0.264 | — | **0.29 ± 0.020** | — | 1.18 ± 0.0019 |
| Quadratic chirp | — | 0.165 ± 0.055 | — | 0.29 ± 0.011 | — | 1.12 ± 0.0014 |
| Logarithmic chirp | — | 0.059 ± 0.027 | — | 0.29 ± 0.006 | — | **1.00 ± 0.0013** |
| Hyperbolic chirp | — | 0.090 ± 0.044 | — | 0.29 ± 0.019 | — | 1.24 ± 0.0015 |

PWC

| Current shape | $\varepsilon_\gamma \times 10^{-2}$ | | $\varepsilon_Z \times 10^{-3}$ | | Normalized clock time | |
|---|---|---|---|---|---|---|
| | Time | FFT | Time | FFT | Time | FFT |
| Pulse | 4.39 ± 0.33 | 4.36 ± 0.35 | 0.258 ± 0.004 | 0.257 ± 0.004 | **1.38 ± 0.0034** | 1.13 ± 0.0048 |
| PRBS | **2.84 ± 0.12** | 2.84 ± 0.12 | **0.256 ± 0.0005** | 0.256 ± 0.0005 | 29.0 ± 0.0029 | **1.00 ± 0.0030** |
| Sawtooth | — | 5.66 ± 0.12 | — | 0.270 ± 0.013 | — | 1.04 ± 0.0031 |
| Linear chirp | — | 3.63 ± 0.72 | — | 0.272 ± 0.015 | — | 1.28 ± 0.0039 |
| Quadratic chirp | — | 2.66 ± 0.19 | — | 0.258 ± 0.002 | — | 1.18 ± 0.0029 |
| Logarithmic chirp | — | **2.56 ± 0.26** | — | 0.256 ± 0.0004 | — | 1.03 ± 0.0083 |
| Hyperbolic chirp | — | 3.97 ± 0.54 | — | **0.256 ± 0.002** | — | 1.12 ± 0.0020 |

The step formulation enables the analysis of digital signals, including the PRBS, which consists of multiple $\pm\Delta i$ pulses with random durations (see Section S1 of the Supplementary Information (SI)).[15] PRBS signals provide broadband frequency content, resulting in greater signal-to-noise ratios compared to single-pulse measurements.[16] For PRBS analysis, the time-domain matrix $\mathbf{B}_{\text{time}}$ can be computed by superposing the individual pulse matrices as given in (12).[17]

*Distribution of relaxation times deconvolution.*—Given an input current, the vector of experimental voltages, $\mathbf{v}_{\text{exp}} = (v_{\text{exp}}(t_1), v_{\text{exp}}(t_2),\ldots,v_{\text{exp}}(t_M))^\top$, can be computed as

$$\mathbf{v}_{\text{exp}} = \mathbf{B}\mathbf{x} + \boldsymbol{\varepsilon} \tag{13}$$

where $\mathbf{B}$ is the discretization matrix for the time data (*i.e.*, $\mathbf{B}_{\text{FFT}}$ or $\mathbf{B}_{\text{time}}$), and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2\, \mathbf{I}_M)$ with $\sigma_\varepsilon$ the noise standard deviation, and $\mathbf{I}_M$ the $M \times M$ identity matrix.

The DRT can be inverted by solving the following quadratic problem:

$$\hat{\mathbf{x}} = \text{argmin}_{\mathbf{x}\geqslant 0}\left(\frac{1}{2}\mathbf{x}^\top\mathbf{Q}\mathbf{x} - \mathbf{q}^\top\mathbf{x}\right) \tag{14}$$

where

$$\mathbf{Q} = \mathbf{B}^\top\mathbf{B} + \lambda\mathbf{M} \tag{15a}$$

$$\mathbf{q} = \mathbf{B}^\top\mathbf{v}_{\text{exp}} \tag{15b}$$

In (15a), $\lambda$ is a positive scalar,[18] and

$$\mathbf{M} = \begin{pmatrix} \alpha & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_{\text{aristo}}^\top\mathbf{L}_{\text{aristo}} \end{pmatrix} \tag{16}$$

with $\alpha$ being a positive scalar and $\mathbf{L}_{\text{aristo}}$ the second-order Aristotelian differentiation matrix. Specifically, $\mathbf{L}_{\text{aristo}}$ depends on the parameter $\sigma_{\text{aristo}}$,[19] and it is defined as

$$\mathbf{L}_{\text{aristo}} = \begin{pmatrix} \sigma_{\text{aristo}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{L} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \sigma_{\text{aristo}} \end{pmatrix} \tag{17}$$

where $\mathbf{L}$ is the second-order differentiation matrix.[20] It should be noted that a large $\sigma_{\text{aristo}}$ penalizes the variability of $\gamma(\log\tau)$ at the boundaries of the discretization interval, where the two boundary

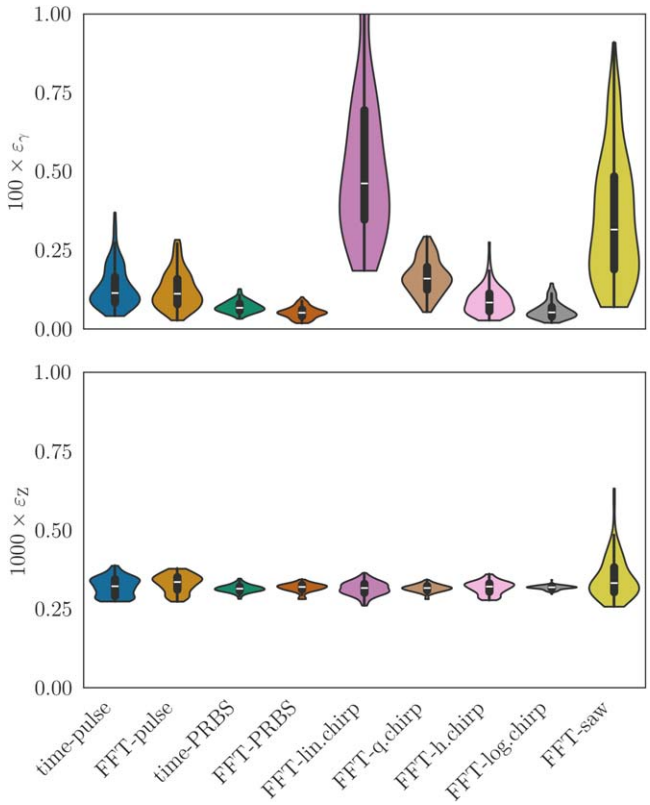


**Figure 2.** For 100 synthetic experiments generated with the 2 × ZARC model, the upper and lower panels show the violin plots of the normalized DRT and impedance residuals, respectively.

values tend to 0 when $\sigma_{\text{aristo}} \to \infty$. Furthermore, setting $\sigma_{\text{aristo}} \neq 0$ renders the matrix $\mathbf{M}$ positive definite, implying that not including this term underlies an improper probability density function.[19]

After recovering $\hat{\boldsymbol{x}}$ by solving (14), we computed the impedance using the standard DRT integral discretization matrix (4). Using a logarithmically spaced frequency vector $\boldsymbol{\omega}_{\log}$ analogous to that used in experimental practice, the impedance is recovered as

$$\hat{\mathbf{Z}} = \mathbf{A}_{\log}\hat{\boldsymbol{x}} \tag{18}$$

where $\mathbf{A}_{\log}$ is the DRT integral discretization matrix (4) constructed using $\boldsymbol{\omega}_{\log}$ and $\log \boldsymbol{\tau} = (\log \tau_1, \log \tau_2, \ldots, \log \tau_N)^{\mathsf{T}}$.[20]

***Implementation.***—To validate our DRT inversion framework, we generated synthetic experimental voltages $\mathbf{v}_{\text{exp}}$ (13) using four distinct families of input signals, including pulse (10) (Fig. S1 a), PRBS (Fig. 1a), sawtooth, and chirps (Section S1 and Fig. 1b for the logarithmic chirp).[16,21] Notably, the sawtooth and chirps cannot be processed using the step approach (11) as linear combinations of pulses cannot approximate either the sawtooth or the chirp.[6] We sampled the time data between 0 and 100 s with a step of 200 $\mu$s using four test circuits: single ZARC, double ZARC, piecewise-constant (PWC), as well as the series combination of a ZARC and a Warburg element[4] to simulate real battery data. The circuit parameters are given in Table S1, with analytical DRTs and impedances provided in Section S1. Moreover, we set $\sigma_\varepsilon/\Delta i = 2.50 \times 10^{-1}\ \Omega$ for the signal-to-noise ratio. The 101-dimensional $\log \boldsymbol{\tau}$ used to discretize (7) was chosen to be logarithmically spaced between $10^{-3}$ s and $10^{3}$ s for the ZARCs and PWC models, and between $1.00 \times 10^{-4}$ s and $1.00 \times 10^{3}$ s for the ZARC +Warburg.

To assess the quality of the recovered EIS DRT, we defined normalized residuals for the DRT and impedance as

$$\varepsilon_\gamma = \frac{\|\boldsymbol{\gamma}_{\text{exact}} - \hat{\boldsymbol{\gamma}}\|^2}{\|\boldsymbol{\gamma}_{\text{exact}}\|^2} \tag{19a}$$

$$\varepsilon_Z = \frac{\|\mathbf{Z}_{\text{exact}} - \hat{\mathbf{Z}}\|^2}{\|\mathbf{Z}_{\text{exact}}\|^2} \tag{19b}$$

where $\boldsymbol{\gamma}_{\text{exact}}$ is the exact DRT at $\log \boldsymbol{\tau}$ (Section S1), $\mathbf{Z}_{\text{exact}}$ is the exact impedance at $\boldsymbol{\omega}_{\log}$ (18),[18] $(\hat{\boldsymbol{\gamma}})_n = (\hat{\boldsymbol{x}})_{n+1}$ for $n = 1, 2, \ldots, N$, and $\hat{\mathbf{Z}}$ is given in (18). The angular frequencies in $\boldsymbol{\omega}_{\log}$ were taken to be logarithmically spaced with ten frequencies per decade between the absolute value of the minimum and maximum angular frequencies obtained by FFTing the time data $\mathbf{t}$ (9). The hyperparameters $\lambda$, $\alpha$, and $\sigma_{\text{aristo}}$ in (15) were obtained by minimizing (19a) within the set $(\log_{10}\lambda, \log_{10}\alpha, \log_{10}\sigma_{\text{aristo}}) \in [-8, 2] \times [-5, 5] \times [-8, 5]$. Exact determination of the optimal parameters was only possible because synthetic impedance data with exact DRT was known. For real experimental data with unknown exact DRT, these three parameters can be obtained using various methods.[18,22] Having assumed $\mathbf{Q}$ to be positive definite, (14) was solved using the MOSEK solver[23] in CVXPY.[24] All reported computations were performed on an Intel(R) Core(TM) i7-11800H CPU (2.30 GHz) with 16 GB RAM. The total clock time included the computation of the $\mathbf{B}$ matrix and the DRT recovery (see Table I).

## Results

Our initial validation focused on comparing our FFT-based approach with the conventional time-domain method using a step current input (Fig. S1 a).[6] The new FFT-based DRT deconvolution method was slightly more accurate in recovering the exact DRT of the 1 × ZARC, 2 × ZARC, and PWC models, and both methods achieved comparable performance in impedance regression. This was validated quantitatively as shown in Table I with the normalized DRT and impedance residuals obtained with 100 synthetic experiments (see (19)). Moreover, Fig. 2 shows the violin plots of the normalized residuals for the 2 × ZARC model, while the recovered DRTs and impedances for this impedance model are shown in Fig. S1 b–d.

For the PRBS signal with $N_{\text{bit}}$ random sequences (Fig. 1a), which represents a generalization of the pulse signal,[16] the FFT approach achieved a slightly improved DRT recovery compared to the time-domain method (Fig. 3a for the 2 × ZARC model), and both methods led to comparable impedance recoveries (Fig. 3c). This finding was confirmed using 100 synthetic voltage signals (see Table I and the violin plots in Fig. 2). The PRBS input proved especially effective at recovering continuous DRTs in both ZARC models. Moreover, the time-domain approach was approximately 27 times more time-consuming, primarily due to calculating $N_{\text{bit}}$ matrices $\mathbf{B}_{\text{time}}$ (see (12) and Section S1). Importantly, the computational time for the FFT approach includes both the construction of the $M \times (N+1)$ matrix $\mathbf{B}_{\text{FFT}}$ with $M = 54$ and $N = 5.00 \times 10^5$ (Section Distribution of relaxation times deconvolution), and the subsequent DRT deconvolution via (14).

A key strength of our FFT-based DRT model lies in its ability to process arbitrary input signals, including chirp signals (as shown in Figs. 1b and 1d for the logarithmic case) and sawtooth functions. This capability significantly extends the limitations of the time-domain method. For illustrative purposes, panels (e)-(j) of Fig. S1 show the DRTs and impedances of the 2 × ZARC and PWC models recovered using the logarithmic chirp. Using 100 synthetic voltages, we found that the logarithmic chirp led to the most accurate DRT and impedance recoveries, see Table I and Fig. 2, for both continuous (1 × ZARC, 2 × ZARC) and discontinuous (PWC) models.

We also analyzed artificial battery data modeled with the 1 × ZARC+Warburg model (Section S1, and Fig. 3b), using the

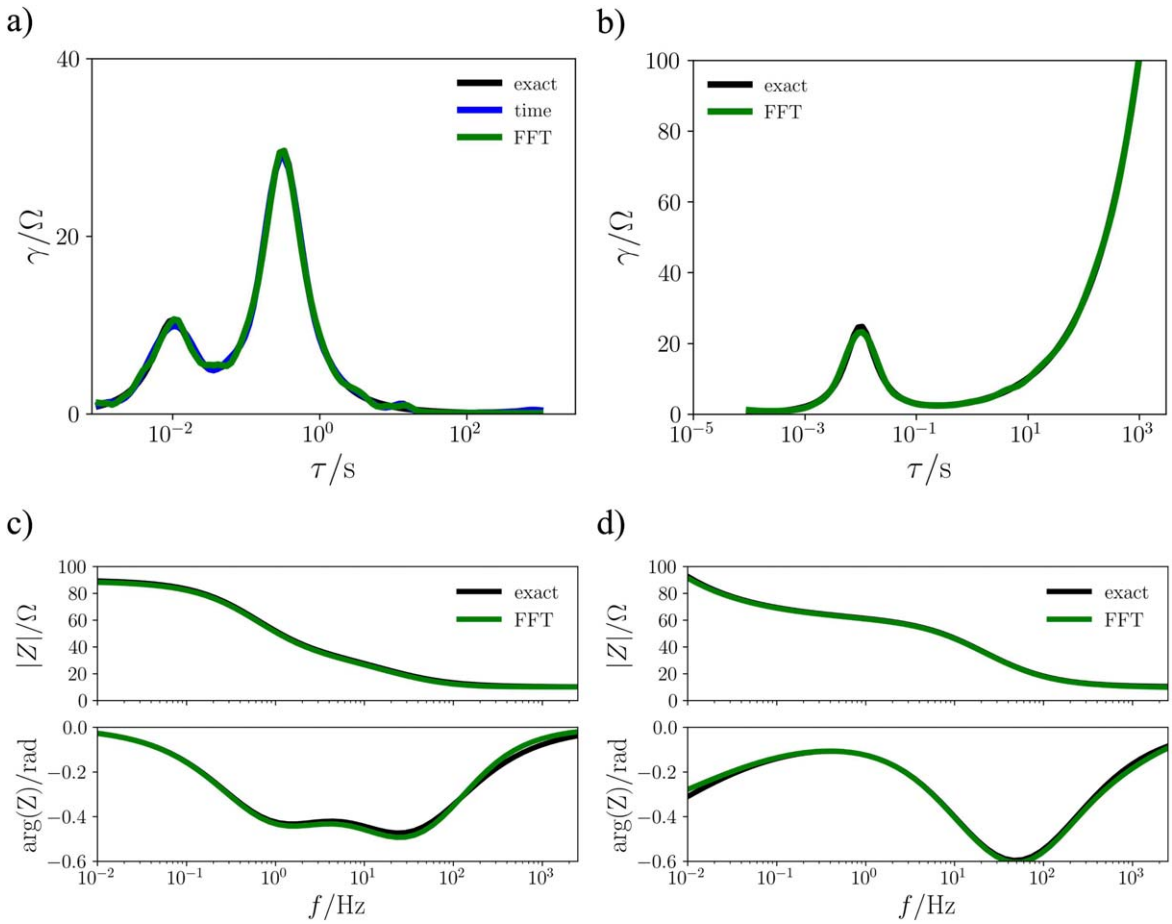


**Figure 3.** For one synthetic experiment generated with the 2 × ZARC model and the PRBS signal, (a) exact and recovered DRTs, and (c) Bode plots of the exact and FFT-recovered impedances. The corresponding plots for one synthetic experiment generated with the ZARC+Warburg impedance are shown in panels (b) and (d).

PRBS signal (Fig. 1a) based on its superior performance (Table I). We employed parameters from Table S1, and derived the exact DRT using the Fuoss and Kirkwood formula[25,26] (Section S1). The FFT method successfully recovered both the exact DRT and impedance, as shown in panels (b) and (d) of Fig. 3, and confirmed with the normalized DRT ($\varepsilon_\gamma = 2.06 \times 10^{-4}$) and impedance ($\varepsilon_Z = 2.53 \times 10^{-4}$) residuals.

## Conclusions

This work presents a novel methodology for simultaneous recovery of impedance and DRT from voltage responses to arbitrary current excitation under a DRT constraint. Moving beyond simple pulse inputs to arbitrary current waveforms, the method significantly expands the scope of time-domain analysis. Through systematic validation against established techniques, we demonstrated that our approach effectively recovers both continuous and discontinuous DRTs and impedances across different input currents, establishing its reliability and accuracy.

Our initial validation using artificial EIS data with known DRT and impedance characteristics yielded promising results, highlighting several valuable research directions. First, this framework could be adapted to other impedance models beyond those studied here. Additionally, validation across real electrochemical devices - including batteries, electrolyzers, and fuel cells - would demonstrate the practical applicability of this approach in experiments. As the method matures, this FFT-based DRT approach could serve as a powerful complement to existing electrochemical analysis tools. Its ability to handle arbitrary inputs while maintaining accuracy makes it particularly valuable for advancing electrochemical diagnostics, deconvolution analysis, and fundamental investigations of complex electrochemical systems.

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